Code Coverage Report

Project: Task Manager Application

Version: 2.0 | Section: J | Batch: 12 | Date: 05-11-2026 | Build under test: `v1.0.1` (Sprint 3)

Authors

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Revision History

Version	Date	Author	Change Summary
1.0	04-11-2026	Student Team	Statement coverage for tag v1.0.
2.0	05-11-2026	Student Team	Statement coverage for v1.0.1.

Approvals

Role	Name	Signature / Date
QA Lead	Tharunyah Sudhakar	
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1. Result

Test set	Statements	Covered	Missed	Statement coverage
v1.0 code, tests as tagged	888	856	32	96.4%
v1.0 code, with Part 3 tests	888	888	0	100.0%
v1.0.1 code, all tests	995	995	0	100.0%

Measured with: coverage.py 7.17.2, over the full unit test suite (155 tests, 155 passed). Line numbers in the gap tables refer to the v1.0 source.

2. Coverage by module

Module	Statements	Covered	Missed	Coverage	Missed lines
app/__init__.py	69	69	0	100.0%	None
app/access.py	12	12	0	100.0%	None
app/admin.py	80	80	0	100.0%	None
app/auth.py	78	78	0	100.0%	None
app/config.py	28	28	0	100.0%	None
app/dashboard.py	18	18	0	100.0%	None
app/db.py	46	46	0	100.0%	None
app/errors.py	40	40	0	100.0%	None
app/notifications.py	117	117	0	100.0%	None

Module	Statements	Covered	Missed	Coverage	Missed lines
app/scheduling.py	109	109	0	100.0%	None
app/search.py	52	52	0	100.0%	None
app/security.py	87	87	0	100.0%	None
app/status.py	4	4	0	100.0%	None
app/tasks.py	124	124	0	100.0%	None
app/utils.py	28	28	0	100.0%	None
app/validators.py	103	103	0	100.0%	None

3. Gaps found in the v1.0 tests and closed in Part 3

Measuring coverage showed which code no test had run. `tests/test_part3_additions.py` now runs it.

Module	Line	Code not run by the v1.0 tests
app/admin.py	106	`raise ValidationError("limit must be an integer") from None`
app/db.py	95	`init_db()`
app/db.py	96	`click.echo(f"Initialised database at {current_app.config['DATABASE']}")`
app/errors.py	72	`return err`
app/notifications.py	57	`if not current_app.config["EMAIL_NOTIFICATIONS_ENABLED"]:`
app/notifications.py	58	`return False`
app/notifications.py	60	`current_app.logger.info("EMAIL to=%s subject=%s", to_address, subject)`
app/notifications.py	61	`return True`
app/notifications.py	106	`continue`
app/notifications.py	118	`interval = app.config["REMINDER_POLL_SECONDS"]`
app/notifications.py	120	`def loop():`
app/notifications.py	121	`while True:`
app/notifications.py	122	`try:`
app/notifications.py	123	`with app.app_context():`
app/notifications.py	124	`count = process_due_reminders()`
app/notifications.py	125	`if count:`
app/notifications.py	126	`app.logger.info("Delivered %d reminder notification(s)", count)`
app/notifications.py	128	`app.logger.exception("Reminder scheduler iteration failed")`
app/notifications.py	129	`time.sleep(interval)`
app/notifications.py	131	`thread = threading.Thread(target=loop, name="reminder-scheduler", daemon=True)`
app/notifications.py	132	`thread.start()`
app/notifications.py	133	`return thread`
app/notifications.py	181	`with transaction() as db:`
app/notifications.py	182	`db.execute("UPDATE notifications SET read_at = ? WHERE user_id = ? AND read_at IS NU LL",`
app/notifications.py	184	`return ok({"message": "All notifications marked as read"})`
app/security.py	87	`with transaction() as tx:`
app/security.py	88	`tx.execute(sql, (now_db(), user_id))`

Module	Line	Code not run by the v1.0 tests
app/tasks.py	180	`return ok({"task": load_task(task_id)})`
app/validators.py	43	`raise ValidationError(f"Name must be at most {MAX_NAME} characters", {"field": "nam e"})`
app/validators.py	78	`raise ValidationError("Description must be text", {"field": "description"})`
app/validators.py	80	`raise ValidationError(f"Description must be at most {MAX_DESCRIPTION} characters",`
app/validators.py	150	`continue`

What these figures do not show

- **Scope:** the Python package `app/`, as in CI. `run.py` and the front-end JavaScript in `app/static/` are not measured here; the front-end is exercised by the browser checks in the System Validation Report.
- **Covered does not mean correct.** The Part 3 tests ran every statement of v1.0 while defects DEF-01 to DEF-05 were still in that code: coverage shows that code ran, not that every input is handled. Sprint 3 therefore added route-wide boundary and fuzz tests.